RAK12002 WisBlock RTC Module Datasheet

Overview

Description

The RAK12002 is a Real-Time Clock module, part of the RAK Wireless WisBlock Series, designed to provide real-time clock capabilities to your WisBlock projects. The RTC chip is based on the RV-3028-C7 from Micro Crystal and uses the I2C interface.

Features

- Built-in 32.768 kHz crystal oscillator
- Counters for seconds, minutes, hours, date, month, year, and day of the week.
- Automatic leap-year correction
- 3.3 V power supply
- I2C Interface
- Programmable Clock output
- Extreme low-current consumption 40 nA
- Super capacitor backup power supply
- Chipset: Micro Crystal RV-3028-C7
- Small size: 10 mm x 10 mm

Specifications

Overview

Mounting

The RAK12002 RTC module can be mounted on slot A, B, C, or D of the WisBlock Base board. Figure 1 shows the mounting mechanism of the RAK12002 on a WisBlock Base board.

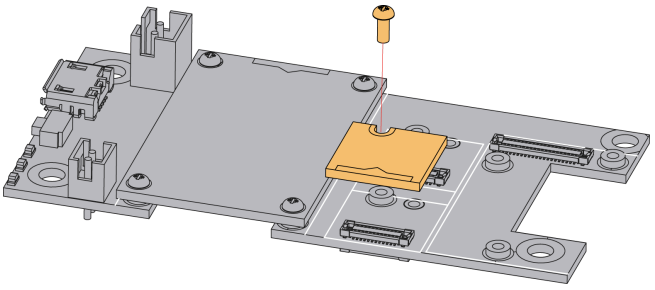


Figure 1: RAK12002 WisBlock RTC Module Mounting

Hardware

The hardware specification is categorized into five parts. It shows the chipset of the module and discusses the pinouts of the module and its corresponding functions and diagrams. It also covers the electrical and mechanical parameters, which include the tabular data of the functionalities and standard values of the RAK12002 WisBlock RTC Module.

Chipset

Vendor	Part number
Micro Crystal	RV-3028-C7

Pin Definition

The RAK12002 WisBlock RTC module includes a standard WisConnector. The WisConnector allows the RAK12002 module to be mounted to a WisBlock Base board. The pin order of the connector and the pinout definition is shown in Figure 2.

> **NOTE**
>
> **I2C** related pins,**INT**, **CLKOUT**, **3V3**, and **GND** are connected to WisConnector.

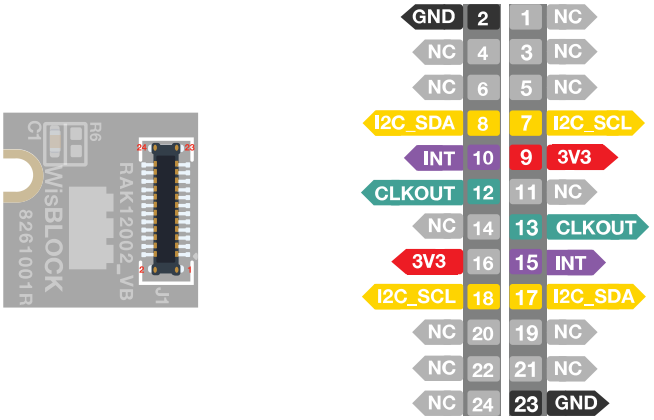


Figure 2: RAK12002 WisBlock RTC Module Pinout

Pin Number	Description	Pin Number	Description
1	NC	2	GND
3	NC	4	NC
5	NC	6	NC
7	I2C_SCL	8	I2C_SDA
9	3V3	10	INT
11	NC	12	CLKOUT
13	CLKOUT	14	NC
15	INT	16	3V3
17	I2C_SDA	18	I2C_SCL
19	NC	20	NC
21	NC	22	NC
23	GND	24	NC

Electrical Characteristics

Symbol	Description	Min.	Nom.	Max.	Unit
VDD	Power Supply Voltage	-	3.3	-	V
VBACKUP	Backup Supply Voltage	-	3.3	-	V
IDD	VDD supply current timekeeping I2C-bus inactive, CLKOUT disabled, average current	-	-	400	nA

Mechanical Characteristic

Board Dimensions

Figure 3 shows the dimensions and the mechanic drawing of the RAK12002 module.

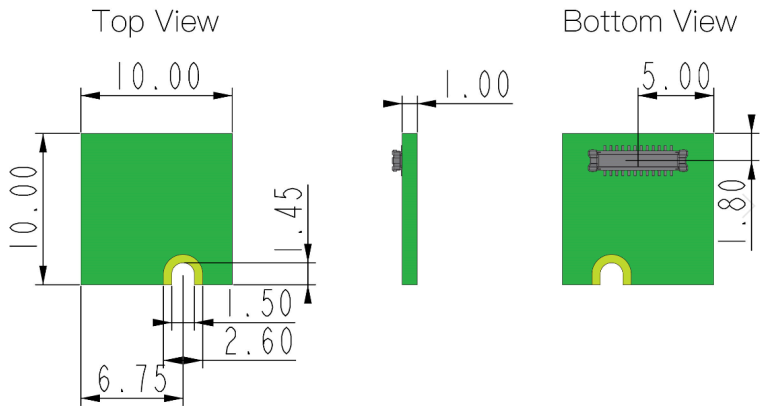


Figure 3: RAK12002 WisBlock RTC Module Mechanic Drawing

WisConnector PCB Layout

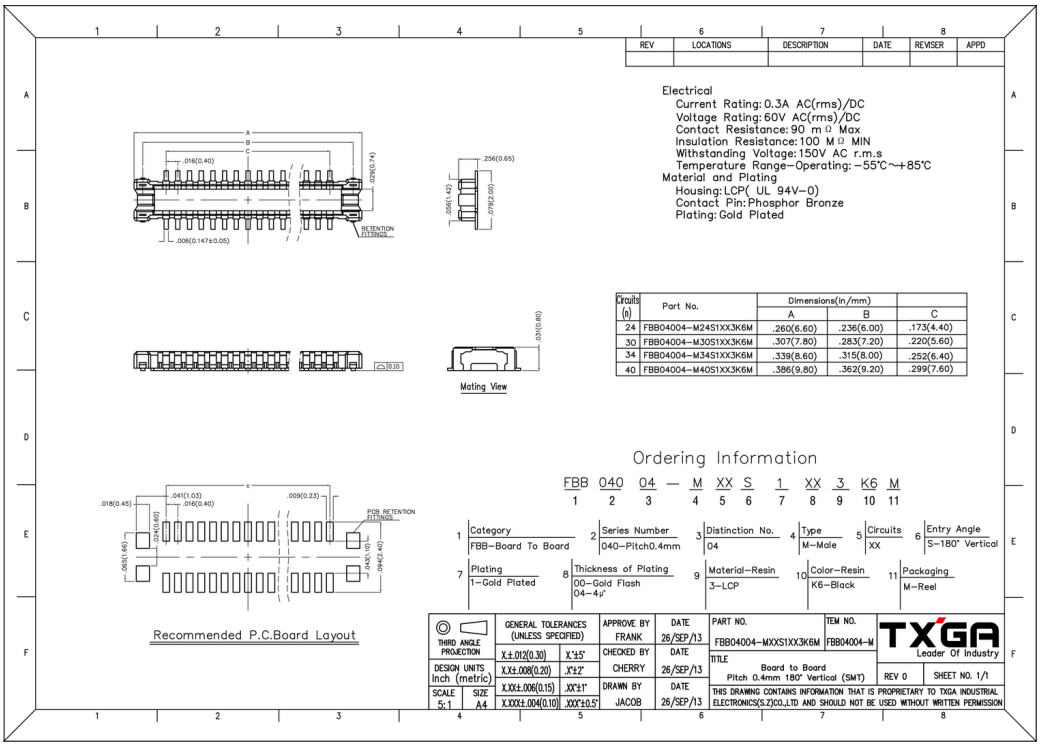


Figure 4: WisConnector PCB Footprint and Recommendations

Schematic Diagram

Figure 5 shows the RAK12002 RTC Module Schematic Diagram. C2 and C3 are supercapacitors, default uses C2, 70 mF/3.3 V, 11 mF/3.3 V for reserved.

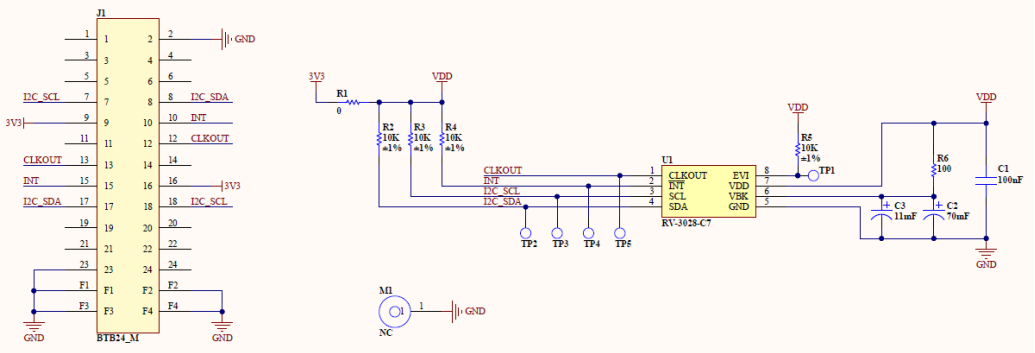


Figure 5: RAK12002 WisBlock RTC Module Schematic